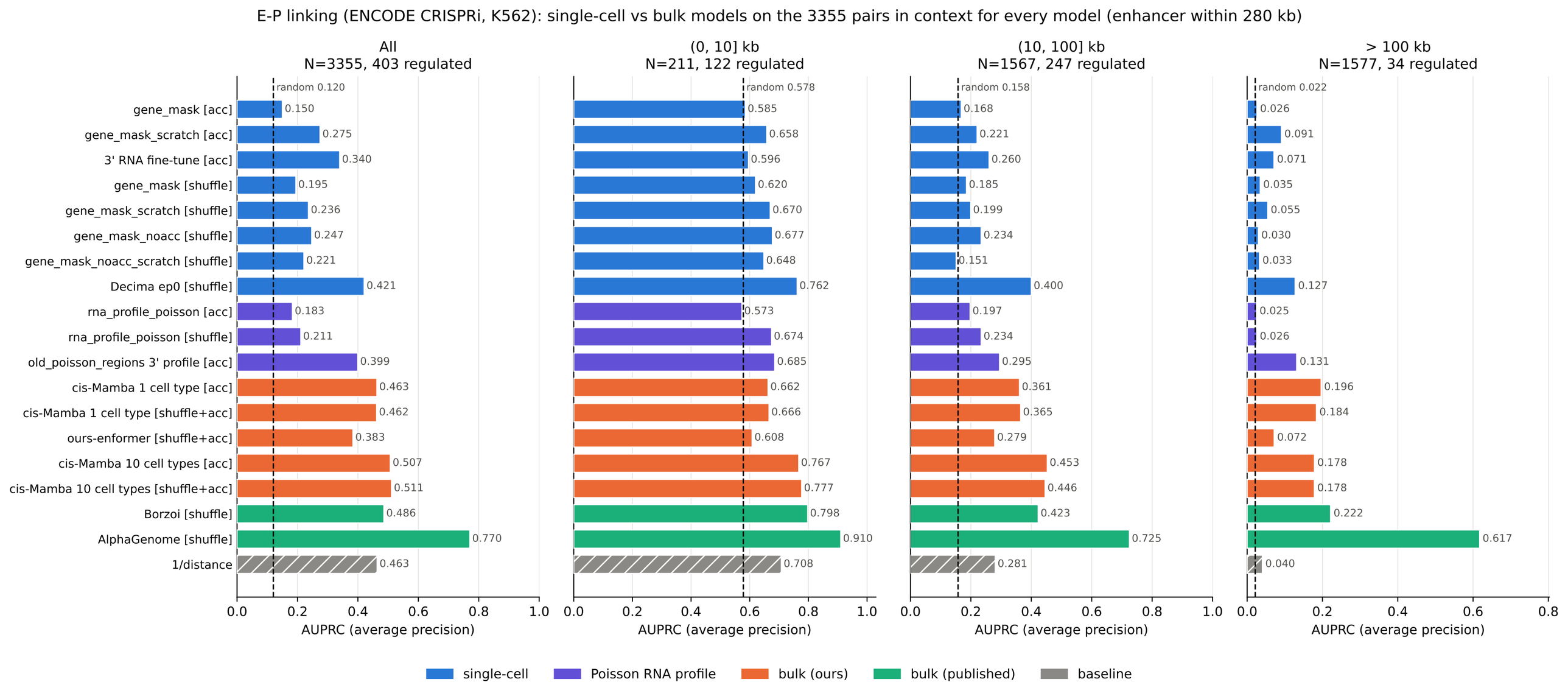
E-P linking (ENCODE CRISPRi, K562): single-cell vs bulk models on the 3355 pairs in context for every model (enhancer within 280 kb)
All
N=3355, 403 regulated
(0, 10] kb
N=211, 122 regulated
(10, 100] kb
N=1567, 247 regulated
> 100 kb
N=1577, 34 regulated
random 0.120
random 0.578
random 0.158
random 0.022
gene_mask [acc] 0.150 0.585 0.168 0.026
gene_mask_scratch [acc] 0.275 0.658 0.221 0.091
3' RNA fine-tune [acc] 0.340 0.596 0.260 0.071
gene_mask [shuffle] 0.195 0.620 0.185 0.035
gene_mask_scratch [shuffle] 0.236 0.670 0.199 0.055
gene_mask_noacc [shuffle] 0.247 0.677 0.234 0.030
gene_mask_noacc_scratch [shuffle] 0.221 0.648 0.151 0.033
Decima ep0 [shuffle] 0.421 0.762 0.400 0.127
rna_profile_poisson [acc] 0.183 0.573 0.197 0.025
rna_profile_poisson [shuffle] 0.211 0.674 0.234 0.026
old_poisson_regions 3' profile [acc] 0.399 0.685 0.295 0.131
cis-Mamba 1 cell type [acc] 0.463 0.662 0.361 0.196
cis-Mamba 1 cell type [shuffle+acc] 0.462 0.666 0.365 0.184
ours-enformer [shuffle+acc] 0.383 0.608 0.279 0.072
cis-Mamba 10 cell types [acc] 0.507 0.767 0.453 0.178
cis-Mamba 10 cell types [shuffle+acc] 0.511 0.777 0.446 0.178
Borzoi [shuffle] 0.486 0.798 0.423 0.222
AlphaGenome [shuffle] 0.770 0.910 0.725 0.617
1/distance 0.463 0.708 0.281 0.040
0.0 0.2 0.4 0.6 0.8 1.0
AUPRC (average precision)
single-cell
Poisson RNA profile
bulk (ours)
bulk (published)
baseline